

Series HFG1E/C



SET-1

प्रश्न-पत्र कोड
Q.P. Code **56/C/1**

रोल नं.
Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

रसायन विज्ञान (सैद्धान्तिक) CHEMISTRY (Theory)

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **35** प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains **23** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **35** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

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General Instructions :

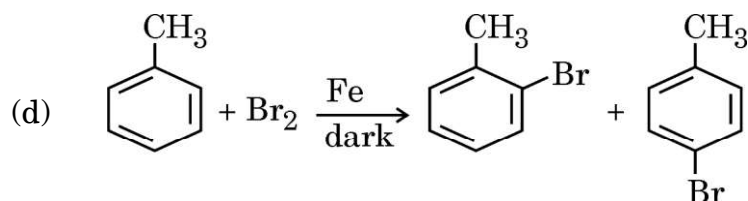
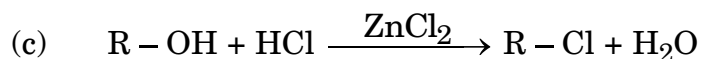
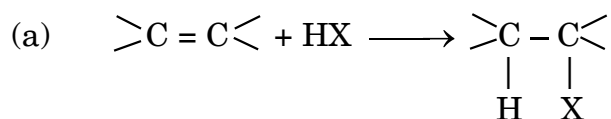
Read the following instructions carefully and strictly follow them :

- (i) This question paper contains **35** questions. **All** questions are **compulsory**.
- (ii) This question paper is divided into **five** Sections – **A, B, C, D** and **E**.
- (iii) In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.
- (iv) In **Section B** – Questions no. **19** to **25** Very Short Answer (VSA) type questions, carrying **2** marks each.
- (v) In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.
- (vi) In **Section D** – Questions no. **31** and **32** are case-based questions carrying **4** marks each.
- (vii) In **Section E** – Questions no. **33** to **35** are Long Answer (LA) type questions carrying **5** marks each.
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 2 questions in Section E.
- (ix) Use of calculators is **not** allowed.

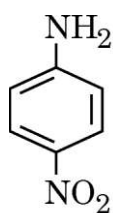
SECTION A

Questions no. **1** to **18** are Multiple Choice (MCQ) type Questions, carrying **1** mark each. 18×1=18

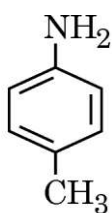
1. Choose the compound which is more acidic than phenol :
 - (a) *o*-nitrophenol
 - (b) ethanol
 - (c) *o*-methylphenol
 - (d) *o*-methoxyphenol
2. Which of the following reactions is a halogenated exchange reaction :



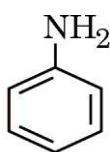
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3. The most common oxidation state for all lanthanoids is :
- (a) + 5 (b) + 2
(c) + 3 (d) + 4
4. A reaction follows second order kinetics. How is the rate of reaction affected if the concentration of the reactant is reduced to half ? Choose the correct value from the following :
- (a) four times (b) eight times
(c) $\frac{1}{4}$ of the original value (d) three times
5. Solutions of two electrolytes X and Y are diluted. Molar conductivity of X increases 25 times whereas that of Y increases 1.5 times. Which one is a stronger electrolyte ?
- (a) X
(b) Y
(c) Both X and Y
(d) None of the above
6. Unit of rate constant for the zero order reaction is :
- (a) s^{-1} (b) $\text{mol}^{-1} \text{L s}^{-1}$
(c) $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$ (d) $\text{mol L}^{-1} \text{s}^{-1}$
7. Three compounds are given below :



I



II



III

The correct decreasing order of their basic strength is :

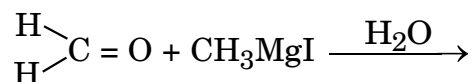
- (a) II > III > I (b) III > II > I
(c) III > I > II (d) I > III > II



8. Which type of isomerism is shown by the complexes $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$?

- (a) Linkage (b) Ionisation
(c) Optical (d) Solvate

9. What would be the major product of the given reaction ?

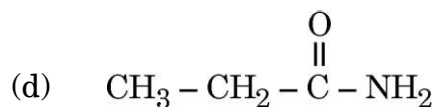
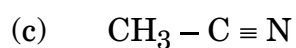
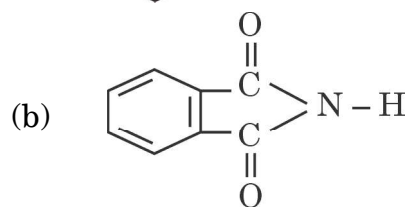
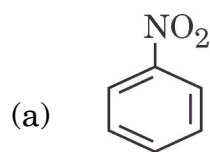


- (a) Ethanal
(b) Propanol
(c) Ethanol
(d) Propanal

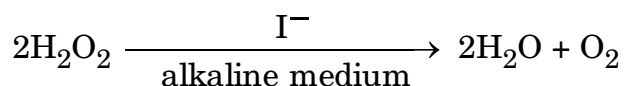
10. Pentan-2-one and Pentan-3-one can be distinguished by :

- (a) Fehling's test
(b) Sodium bicarbonate test
(c) Tollens' test
(d) Iodoform test

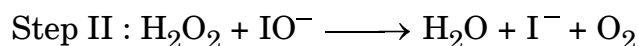
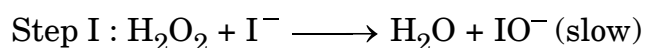
11. Hoffmann Bromamide Degradation reaction is given by :



12. Given below is the decomposition of hydrogen peroxide in alkaline medium, which is catalysed by iodide ions :



The above reaction takes place in two steps :



Molecularity of Step I and Step II is :

- (a) Step I – 2, Step II – 2
(b) Step I – 1, Step II – 2
(c) Step I – 2, Step II – 1
(d) Step I – 3, Step II – 1
13. Oxidation state of central metal atom in the given complex is :
 $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_3$
- (a) + 2 (b) + 3
(c) + 1 (d) + 4
14. A galvanic cell can behave as an electrolytic cell when :
- (a) $E_{\text{cell}} = E_{\text{ext}}$ (b) $E_{\text{cell}} > E_{\text{ext}}$
(c) $E_{\text{cell}} = 0$ (d) $E_{\text{ext}} > E_{\text{cell}}$

For Questions number 15 to 18, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.



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15. *Assertion (A)* : Phenol gives 2,4,6-trinitrophenol on treatment with concentrated HNO_3 and concentrated H_2SO_4 .

Reason (R) : – OH group in phenol is *m*-directing.

16. *Assertion (A)* : D (+)-Glucose is dextrorotatory.

Reason (R) : Symbol 'D' represents its dextrorotatory nature.

17. *Assertion (A)* : Zn, Cd and Hg are not regarded as transition elements.

Reason (R) : Zn, Cd and Hg do not have partially filled d-orbitals in their ground state or in any one of their common oxidation states.

18. *Assertion (A)* : Benzene diazonium salt is stable and can be easily stored.

Reason (R) : Benzene diazonium chloride decomposes easily.

SECTION B

19. The rate of a reaction quadruples when the temperature changes from 293 K to 313 K. Calculate the energy of activation of the reaction, assuming that it does not change with temperature. 2

($\log 2 = 0.30$, $\log 4 = 0.60$) [$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

20. (a) (i) Name any two bases which are common to both DNA and RNA.

(ii) Which vitamin deficiency causes :

(1) Bone deformities in children ?

(2) Pernicious anaemia ? 2

OR

(b) (i) Why do amino acids show amphoteric behaviour ?

(ii) What happens when D-Glucose is treated with hydroxylamine ? 2



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21. (a) (i) Write the structure of 1-chloro-4-ethylcyclohexane.
(ii) Why is sulphuric acid not used during the reactions of alcohols with KI ? 2

OR

- (b) (i) Arrange the following in increasing order of their boiling points :
1-chloropropane, 2-chloropropane, 1-chlorobutane
(ii) What is an ambident nucleophile ? Give one example. 2
22. (a) Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ coloured ?
(b) Write IUPAC name of the given complex :
 $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ 2
23. (a) Why are iron pipes usually coated with zinc ?
(b) Why does mercury cell give a constant voltage throughout its life ? 2
24. (a) Write the mathematical relation between rate constant and half-life period of a zero order reaction.
(b) Define Pseudo first order reaction with an example. 2
25. (a) Why is pK_a of ClCH_2COOH lower than the pK_a value of CH_3COOH ?
(b) Write the chemical equation for Hell-Volhard-Zelinsky reaction. 2

SECTION C

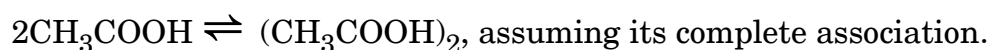
26. Write the chemical equation for the following : 1+1+1
- (a) Hydration of propene in presence of an acid.
(b) Reaction between Ethyl bromide and $\text{C}_2\text{H}_5\text{ONa}$.
(c) Reaction between Dimethyl ether and Hydrogen iodide.



27. Using valence bond theory, predict : 1+1+1

- (a) Hybridisation of central metal atom of the complex $[\text{Co}(\text{NH}_3)_6]^{3+}$.
- (b) Its shape and magnetic behaviour.
- (c) Whether it is a high spin or a low spin complex.
[Atomic number : Co = 27]

28. (a) Find the value of van't Hoff factor for acetic acid in benzene as per the given equation :



- (b) Osmotic pressure of a solution containing 3.5 g of dissolved protein in 0.05 L of a solution is 0.035 atm at 310 K. Calculate the molar mass of the protein. ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$) 1+2

29. Answer any **three** of the following questions : 1+1+1

Give reasons for the following :

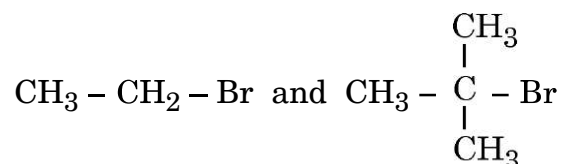
- (a) Reduction with iron scrap and HCl is preferred for the preparation of amines from nitro compounds.
- (b) Lower aliphatic amines are soluble in water.
- (c) Pyridine is used in the acylation reaction of amines.
- (d) Although amino group is *o*- and *p*-directing in aromatic electrophilic substitution reactions, aniline on nitration gives a substantial amount of *m*-nitroaniline.

30. (a) (i) Why do tertiary alkyl halides undergo $\text{S}_{\text{N}}1$ reaction at a faster rate ?
(ii) Define Enantiomers.
(iii) Why is chloroform stored in dark coloured air tight bottles ? 1+1+1

OR



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- (b) (i) Write the major alkene that would be formed by dehydrohalogenation of 2-Bromopentane.
- (ii) Which would undergo S_N2 reaction at a faster rate and why ?



- (iii) Why is chlorobenzene less reactive towards nucleophilic substitution reaction ? $1 + \left(\frac{1}{2} + \frac{1}{2}\right) + 1$

SECTION D

The following questions are case-based questions. Read the case carefully and answer the questions that follow :

- 31.** Carbohydrates are the major components of all living organisms. Sugars are carbohydrates. The major types of sugars include monosaccharides and disaccharides. The main difference between monosaccharides, disaccharides and polysaccharides is that monosaccharides are monomer of sugars and disaccharides are composed of two monomers, whereas polysaccharides are composed of a large number of monomers. Monosaccharides are single sugar molecules which act as the building blocks of disaccharides and polysaccharides. Disaccharides are also simple sugars. Disaccharides are classified into two groups according to their reducing strength : Reducing and Non-reducing sugars. When a polymer is formed from a monomer, a condensation reaction occurs that forms a glycosidic bond and water molecule is lost. Starch, glycogen and cellulose are examples of polysaccharides. Starch is found in many parts of plant cell and consists of amylose and amylopectin. Glycogen is the major carbohydrate storage product found in humans. It is present in liver, muscles and brain.

Cellulose is the most abundant organic molecule on Earth. It makes up around 50% of all organic carbon.



Answer the following questions :

- (a) Name the linkage which connects monosaccharide units in polysaccharides.
- (b) Carbohydrates are classified on the basis of their behaviour on hydrolysis. Write the hydrolysis products of sucrose.
- (c) Write two differences between Amylose and Amylopectin.

OR

- (c) (i) What are reducing sugars ?
- (ii) Sucrose is dextrorotatory but the mixture obtained after hydrolysis is laevorotatory. Why ? 1+1+2

- 32.** Raoult's law for volatile liquids states that the partial vapour pressure of each component in the solution is directly proportional to its mole fraction, whereas for a non-volatile solute, it states that the vapour pressure of a solution of a non-volatile solute is equal to the vapour pressure of the pure solvent at that temperature multiplied by its mole fraction. Two liquids A and B are mixed with each other to form a solution, the vapour phase consists of both components of the solution. Once the components in the solution have reached equilibrium, the total vapour pressure of the solution can be determined by combining Raoult's law with Dalton's law of partial pressures. If a non-volatile solute B is dissolved into a solvent A to form a solution, the vapour pressure of the solution will be lower than that of the pure solvent. The solutions which obey Raoult's law over the entire range of concentration are ideal solutions, whereas the solutions for which vapour pressure is either higher or lower than that predicted by Raoult's law are called non-ideal solutions. Non-ideal solutions are identified by determining the strength of the intermolecular forces between the different molecules in that particular solution. They can either show positive or negative deviation from Raoult's law depending on whether the A – B interactions in solution are stronger or weaker than A – A and B – B interactions.

Answer the following questions :

- (a) 20 mL of a liquid A was mixed with 20 mL of liquid B. The volume of resulting solution was found to be less than 40 mL. What do you conclude from the above data ?



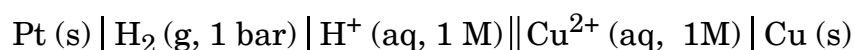
- (b) Which of the following show positive deviation from Raoult's law ?
Carbon disulphide and Acetone; Phenol and Aniline; Ethanol and Acetone
- (c) The vapour pressure of a solution of glucose in water is 750 mm Hg at 100°C. Calculate the mole fraction of solute.
(Vapour pressure of water at 373 K = 760 mm Hg)

OR

- (c) The boiling point of solution increases when 1 mol of NaCl is added to 1 litre of water while addition of 1 mol of methanol to one litre of water decreases its boiling point. Explain the above observations. 1+1+2

SECTION E

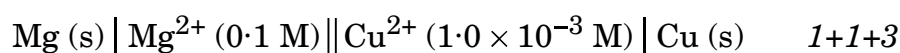
33. (a) (i) A cell and its emf is given below :



emf of the cell = + 0.34 V.

Write the reduction half-reaction at cathode.

- (ii) How is standard Gibbs energy for a reaction related to equilibrium constant ?
- (iii) Calculate emf of the given cell :

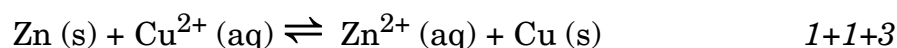


Given : $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = + 0.34 \text{ V}$, $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = - 2.37 \text{ V}$

(log 100 = 2)

OR

- (b) (i) State Kohlrausch's law of independent migration of ions.
- (ii) How much electricity in terms of Faraday is required to produce 40 g of Al from molten Al_2O_3 ?
(Given : atomic mass of Al = 27 u)
- (iii) Calculate log K_c for the following reaction at 298 K :



Given : $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = - 0.76 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = + 0.34 \text{ V}$



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34. (a) Compound A undergoes Rosenmund reduction to give compound B with molecular formula C_7H_6O . Compound B does not give Fehling's test but reacts with conc. NaOH to give C and D.

Identify A, B, C and D and write all the reactions involved.

Write one chemical test to distinguish between compound B and propanone.

5

OR

- (b) Compound A with molecular formula (C_2H_6O) on oxidation by PCC gives compound B, which on treatment with dilute alkali forms compound C which is a β -hydroxy aldehyde. B on oxidation by potassium permanganate forms C. Identify A, B, C and D and write all the chemical equations involved.

5

35. Answer the following questions :

1+1+1+1+1

- (a) The chemistry of the actinoids is more complex as compared to lanthanoids. Why ?
- (b) Why is E° for Mn^{3+}/Mn^{2+} redox couple more positive ?
- (c) Why do transition metals form large numbers of complex compounds ?
- (d) How does acidified potassium permanganate solution react with Fe^{2+} ions ? Write ionic equation.
- (e) Calculate the 'spin only' magnetic moment of a divalent ion of a metal M in aqueous solution. The atomic number of the metal M is 25.

